

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,collect(DISTINCT git_commit.sha) AS commitHashes\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n ,max(git_commit.sha) AS maxCommitSha\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

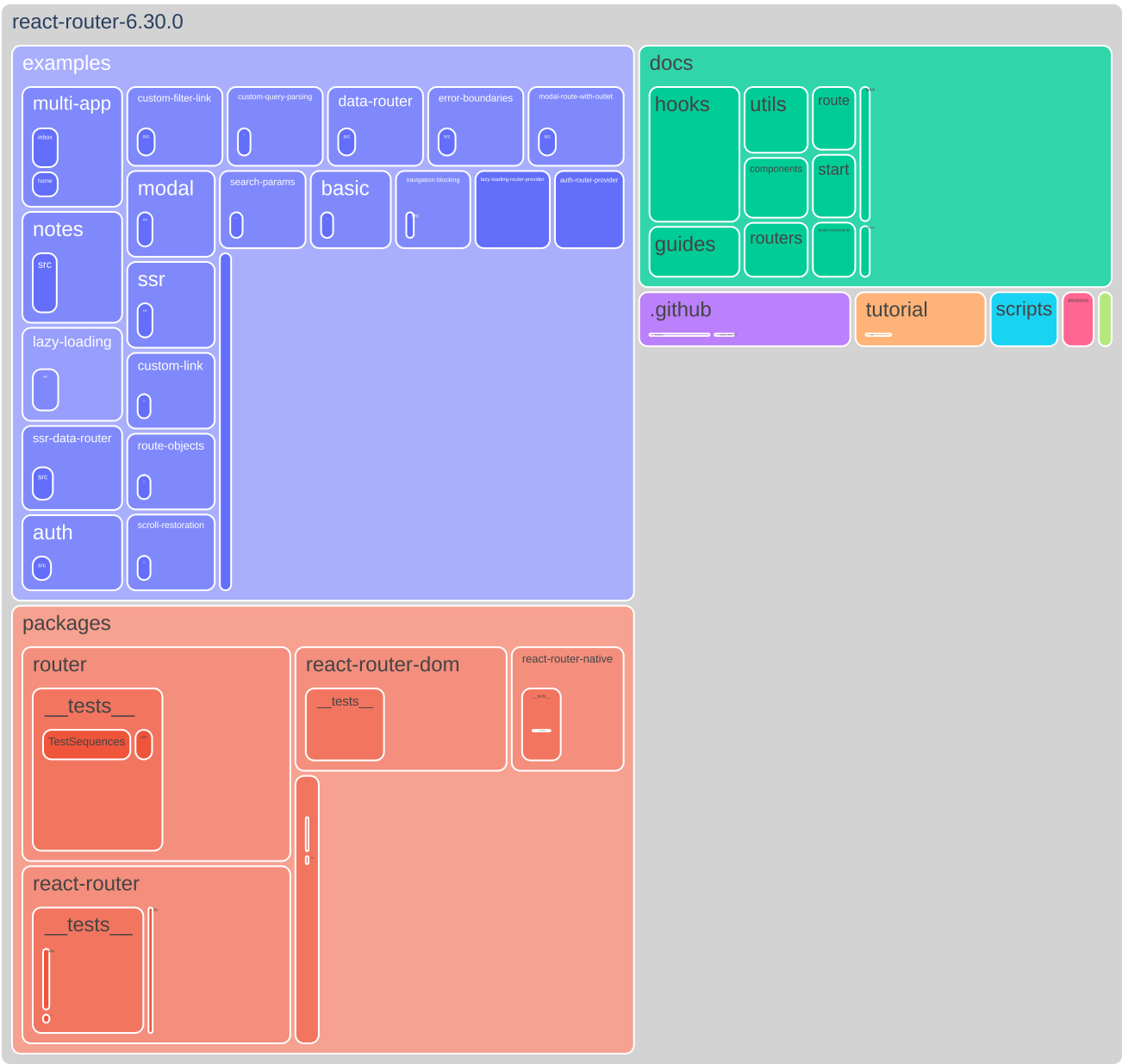
Data Preview

	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	83.000000	83.000000	83.000000	83.000000	83.000000	83.000000
mean	25.301205	19.650602	129.975904	485.975904	889.722892	486.879518
std	77.772453	46.179246	291.778607	277.542657	386.873122	277.949478
min	1.000000	2.000000	3.000000	68.000000	97.000000	67.000000
25%	4.000000	4.000000	11.000000	216.000000	614.000000	215.000000
50%	10.000000	6.000000	26.000000	601.000000	969.000000	600.000000
75%	13.000000	13.500000	83.000000	700.000000	1293.000000	699.000000
max	633.000000	343.000000	1989.000000	1334.000000	1390.000000	1333.000000

	directoryPath	directoryParentPath	directoryName	fileCount	firstFile	mostF
0	react-router-6.30.0/examples/custom-query-parsing/types	react-router-6.30.0/examples/custom-query-parsing	types	1	react-router-6.30.0/examples/custom-query-parsing/types/jsurl.d.ts	
1	react-router-6.30.0/packages/react-router-dom-v5-compat/lib	react-router-6.30.0/packages/react-router-dom-v5-compat	lib	1	react-router-6.30.0/packages/react-router-dom-v5-compat/lib/components.tsx	
2	react-router-6.30.0/packages/react-router-dom/__tests__/polyfills	react-router-6.30.0/packages/react-router-dom/__tests__	polyfills	1	react-router-6.30.0/packages/react-router-dom/__tests__/polyfills/drop-FormData-submitter.ts	
3	react-router-6.30.0/packages/react-router/__tests__/_snapshots__	react-router-6.30.0/packages/react-router/__tests__	__snapshots__	1	react-router-6.30.0/packages/react-router/__tests__/_snapshots__/route-matching-test.tsx.snap	
4	react-router-6.30.0/patches	react-router-6.30.0	patches	1	react-router-6.30.0/patches/@changesets_get-dependents-graph@1.3.6.patch	
5	react-router-6.30.0/.changeset	react-router-6.30.0	.changeset	2	react-router-6.30.0/.changeset/README.md	
6	react-router-6.30.0/docs/upgrading	react-router-6.30.0/docs	upgrading	2	react-router-6.30.0/docs/upgrading/future.md	
7	react-router-6.30.0/examples/lazy-loading/src/pages	react-router-6.30.0/examples/lazy-loading/src	pages	2	react-router-6.30.0/examples/lazy-loading/src/pages/About.tsx	
8	react-router-6.30.0/.github/ISSUE_TEMPLATE	react-router-6.30.0/.github	ISSUE_TEMPLATE	3	react-router-6.30.0/.github/ISSUE_TEMPLATE/bug_report.yml	
9	react-router-6.30.0/examples/navigation-blocking/src	react-router-6.30.0/examples/navigation-blocking	src	3	react-router-6.30.0/examples/navigation-blocking/src/app.tsx	
10	react-router-6.30.0/packages/react-router-dom-v5-compat/__tests__	react-router-6.30.0/packages/react-router-dom-v5-compat	__tests__	3	react-router-6.30.0/packages/react-router-dom-v5-compat/__tests__/compat-router-test.tsx	
11	react-router-6.30.0/decisions	react-router-6.30.0	decisions	4	react-router-6.30.0/decisions/0001-use-blocker.md	
12	react-router-6.30.0/examples/basic/src	react-router-6.30.0/examples/basic	src	4	react-router-6.30.0/examples/basic/src/App.tsx	
13	react-router-6.30.0/examples/custom-link/src	react-router-6.30.0/examples/custom-link	src	4	react-router-6.30.0/examples/custom-link/src/App.tsx	
14	react-router-6.30.0/examples/custom-query-parsing/src	react-router-6.30.0/examples/custom-query-parsing	src	4	react-router-6.30.0/examples/custom-query-parsing/src/App.tsx	
15	react-router-6.30.0/examples/multi-app/home	react-router-6.30.0/examples/multi-app	home	4	react-router-6.30.0/examples/multi-app/home/App.jsx	
16	react-router-6.30.0/examples/route-objects/src	react-router-6.30.0/examples/route-objects	src	4	react-router-6.30.0/examples/route-objects/src/App.tsx	
17	react-router-6.30.0/examples/scroll-restoration/src	react-router-6.30.0/examples/scroll-restoration	src	4	react-router-6.30.0/examples/scroll-restoration/src/app.tsx	
18	react-router-6.30.0/examples/search-params/src	react-router-6.30.0/examples/search-params	src	4	react-router-6.30.0/examples/search-params/src/App.tsx	
19	react-router-6.30.0/packages/react-router-native/__tests__/_snapshots__	react-router-6.30.0/packages/react-router-native/__tests__	__snapshots__	4	react-router-6.30.0/packages/react-router-native/__tests__/_snapshots__/deep-linking-test.tsx.snap	
20	react-router-6.30.0/packages/react-router/lib	react-router-6.30.0/packages/react-router	lib	4	react-router-6.30.0/packages/react-router/lib/components.tsx	
21	react-router-6.30.0/packages/router/__tests__/_utils__	react-router-6.30.0/packages/router/__tests__	utils	4	react-router-6.30.0/packages/router/__tests__/_utils__/custom-matchers.ts	
22	react-router-6.30.0/tutorial/src	react-router-6.30.0/tutorial	src	4	react-router-6.30.0/tutorial/src/App.jsx	
23	react-router-6.30.0/docs/fetch	react-router-6.30.0/docs	fetch	5	react-router-6.30.0/docs/fetch/index.md	
24	react-router-6.30.0/examples/auth/src	react-router-6.30.0/examples/auth	src	5	react-router-6.30.0/examples/auth/src/App.tsx	
25	react-router-6.30.0/examples/custom-filter-link/src	react-router-6.30.0/examples/custom-filter-link	src	5	react-router-6.30.0/examples/custom-filter-link/src/App.tsx	
26	react-router-6.30.0/examples/data-router/src	react-router-6.30.0/examples/data-router	src	5	react-router-6.30.0/examples/data-router/src/app.tsx	
27	react-router-6.30.0/examples/error-boundaries/src	react-router-6.30.0/examples/error-boundaries	src	5	react-router-6.30.0/examples/error-boundaries/src/app.tsx	
28	react-router-6.30.0/examples/modal-route-with-outlet/src	react-router-6.30.0/examples/modal-route-with-outlet	src	5	react-router-6.30.0/examples/modal-route-with-outlet/src/App.tsx	
29	react-router-6.30.0/examples/modal/src	react-router-6.30.0/examples/modal	src	5	react-router-6.30.0/examples/modal/src/App.tsx	

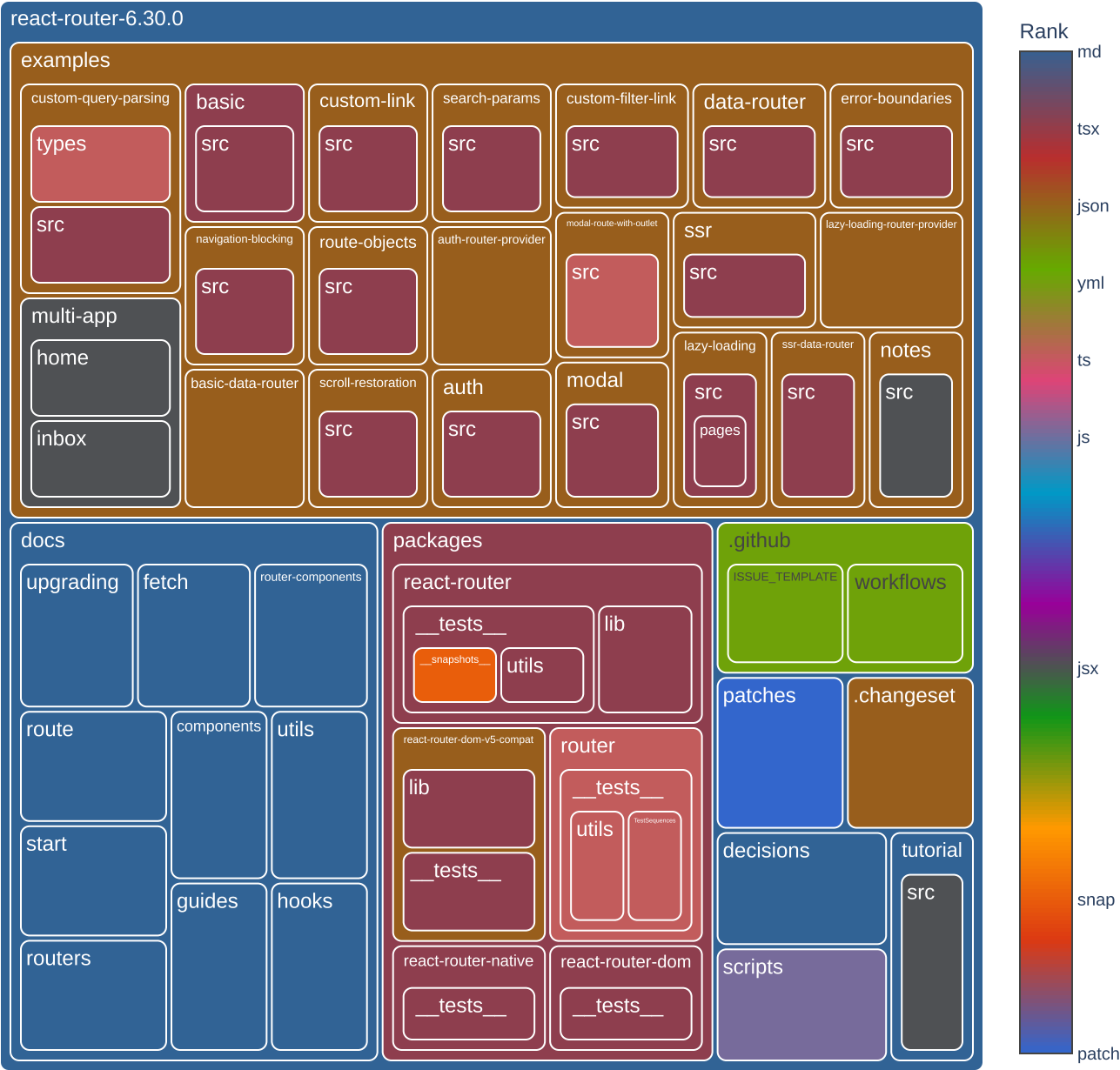
Number of files per directory

Directories and their file count



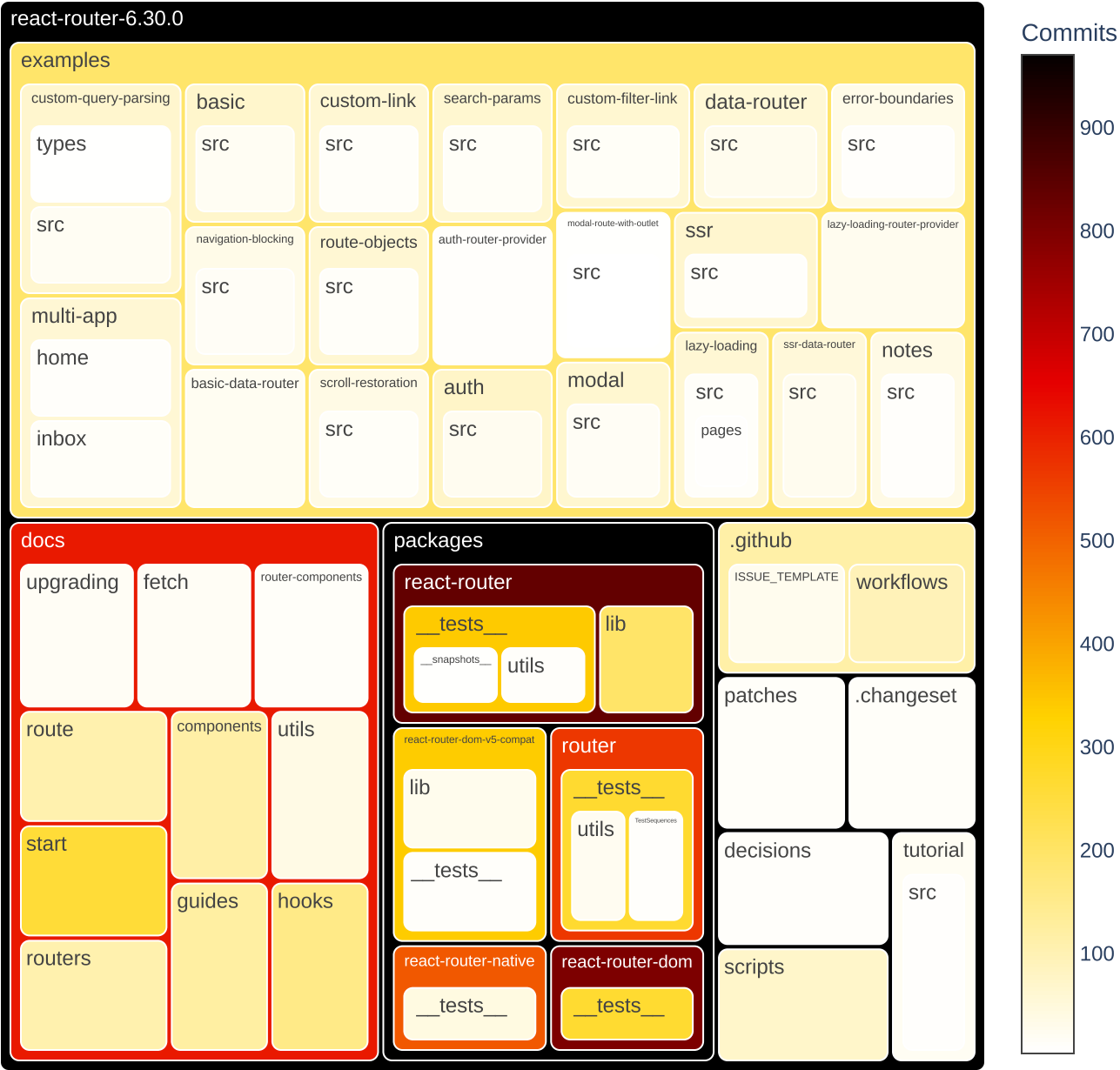
Most frequent file extension per directory

Most frequent file extension per directory



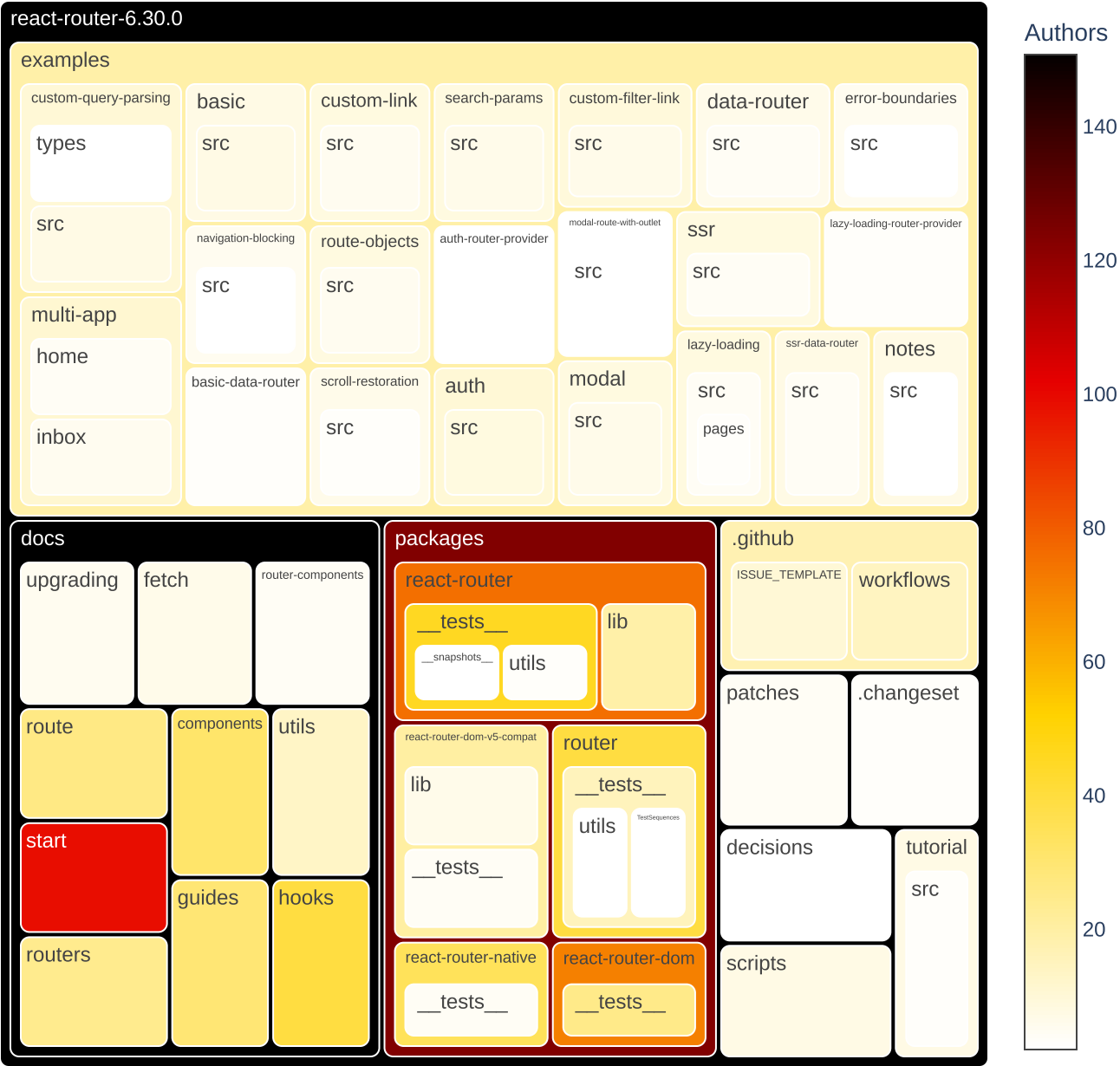
Number of commits per directory

Number of git commits



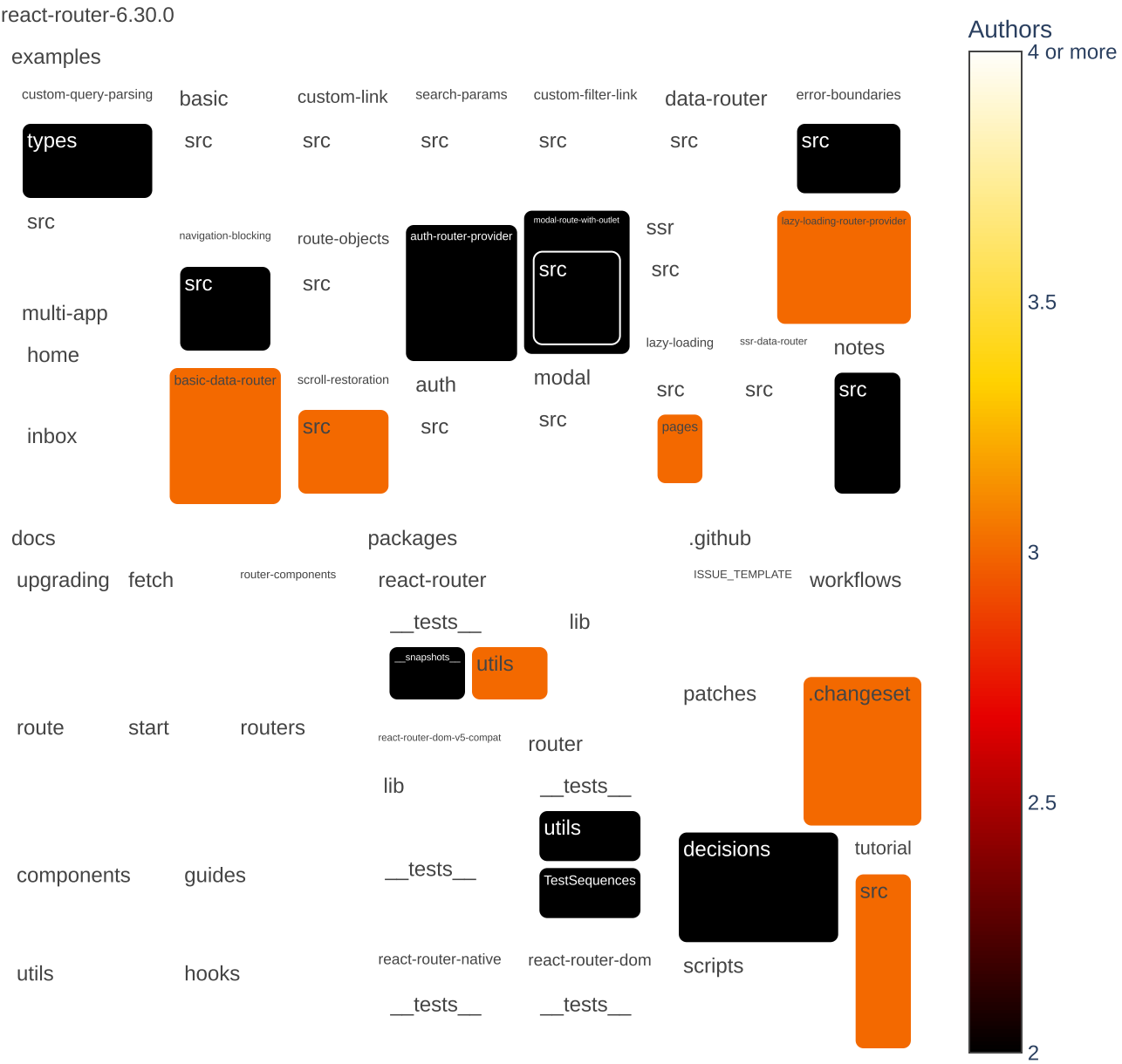
Number of distinct authors per directory

Number of distinct commit authors



Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)



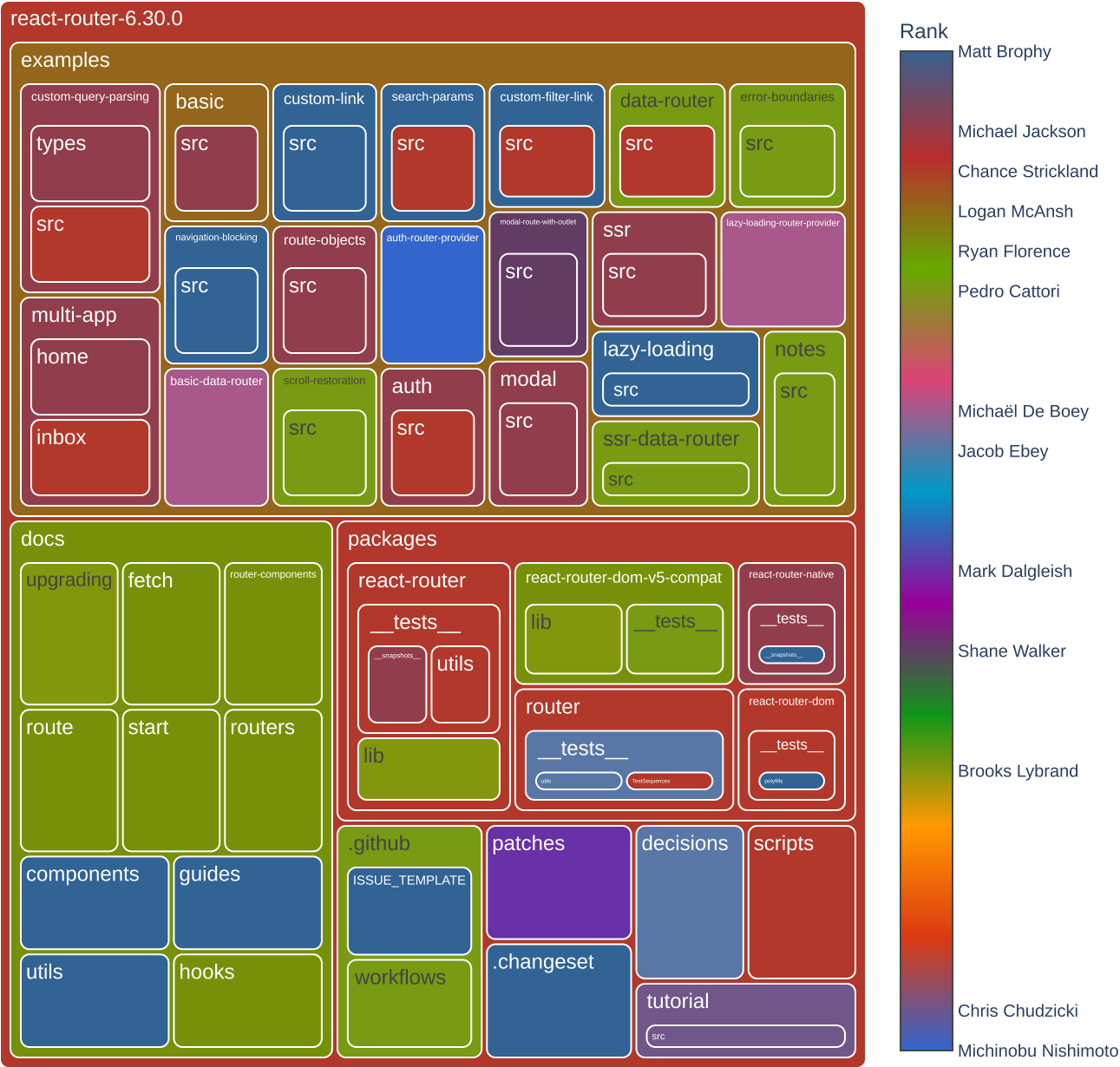
Main author per directory

Main authors with highest number of commits



Second author per directory

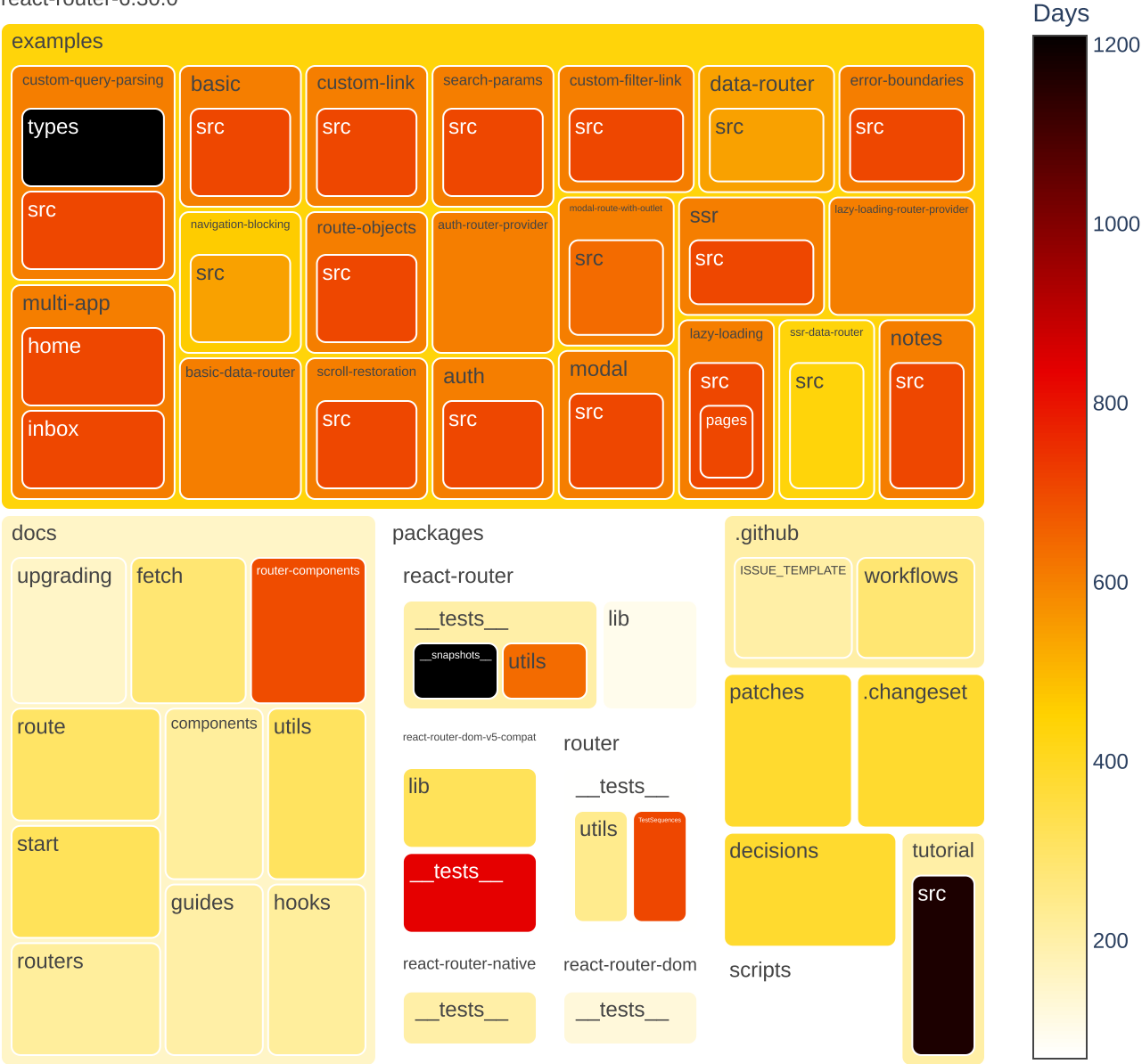
Second author with the second highest number of commits



Days since last commit per directory

Days since last commit

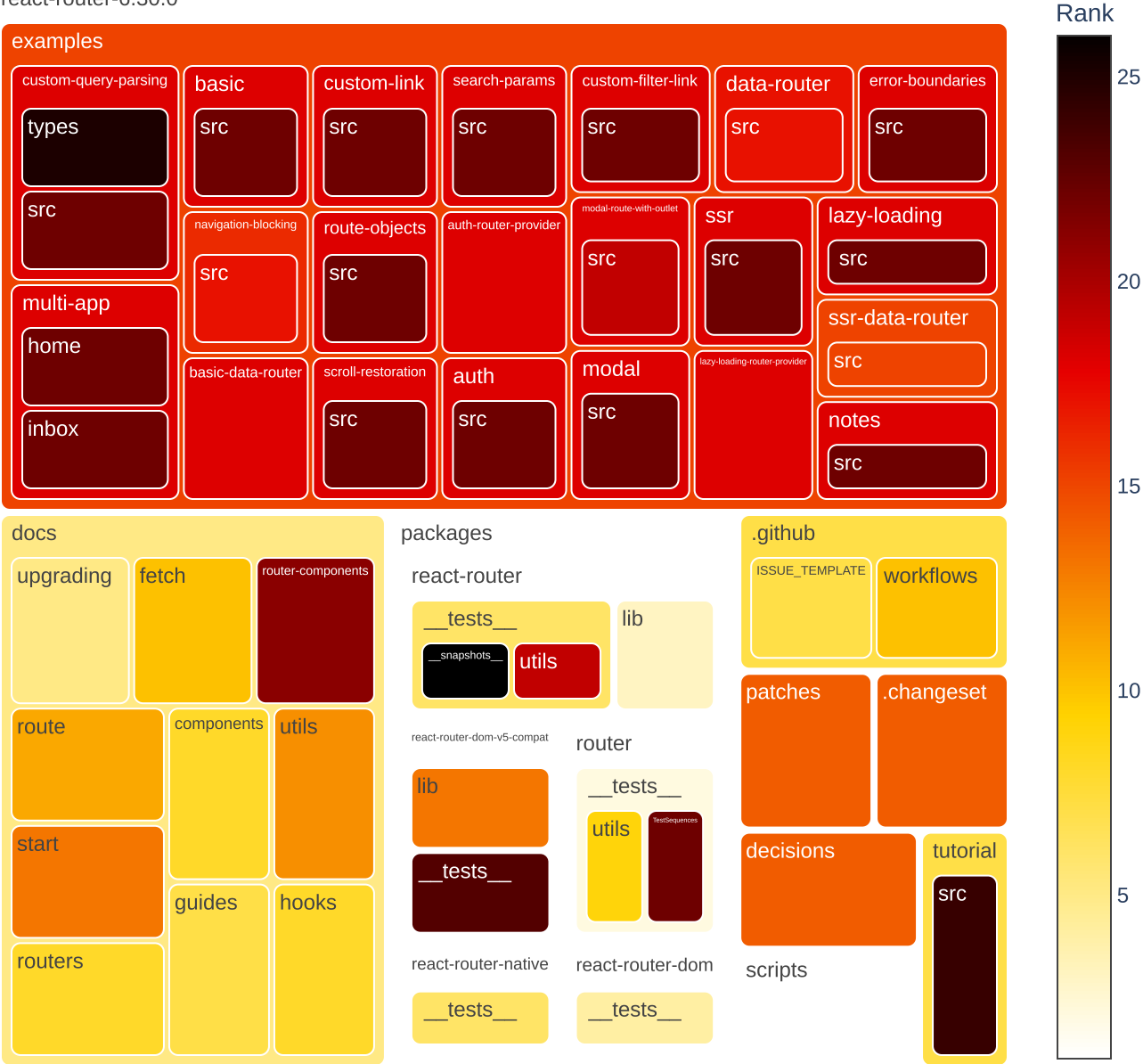
react-router-6.30.0



Days since last commit per directory (ranked)

Rank of days since last commit

react-router-6.30.0



Days since last file creation per directory

Days since last file creation

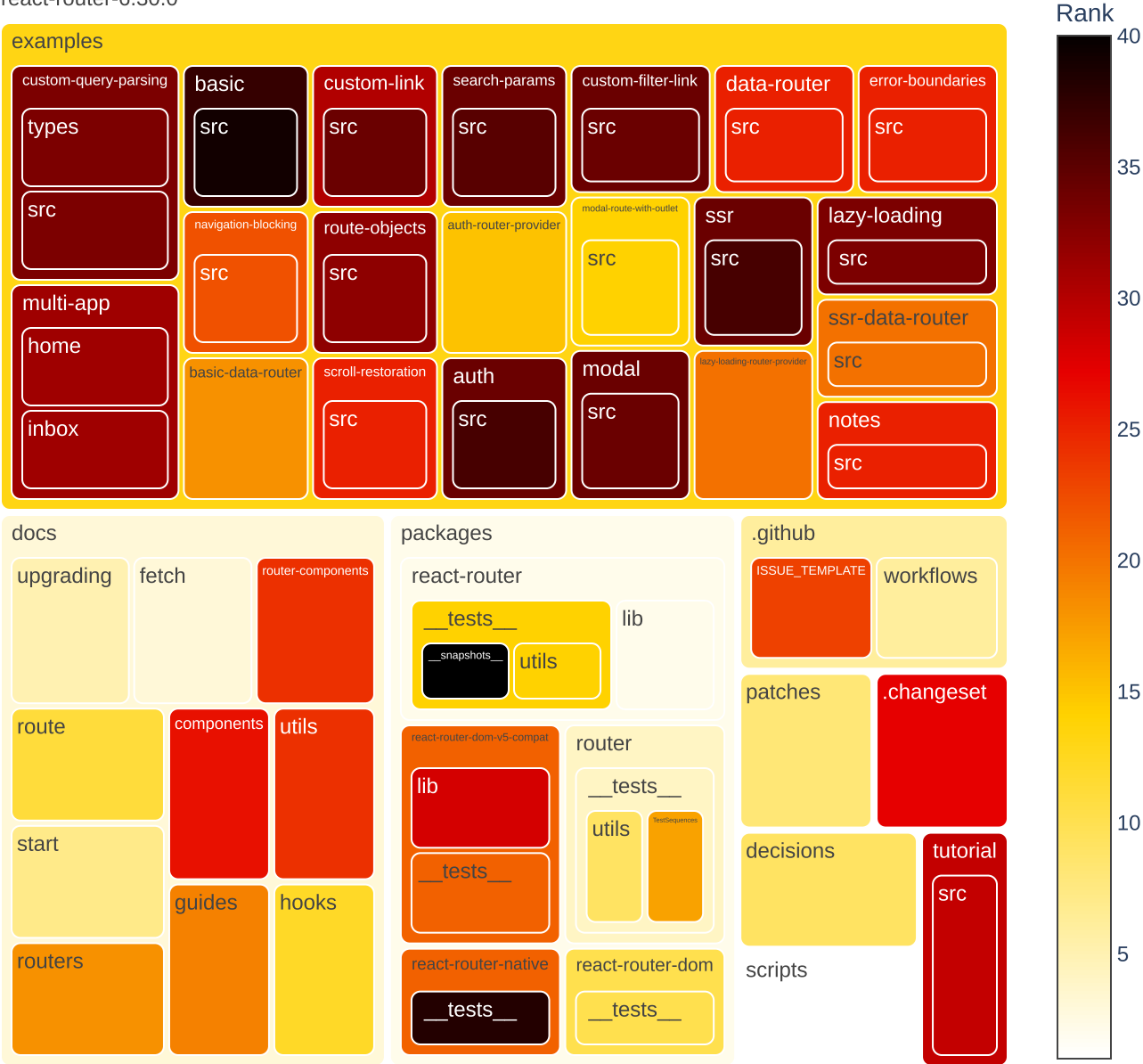
react-router-6.30.0



Days since last file creation per directory (ranked)

Rank of days since last file creation

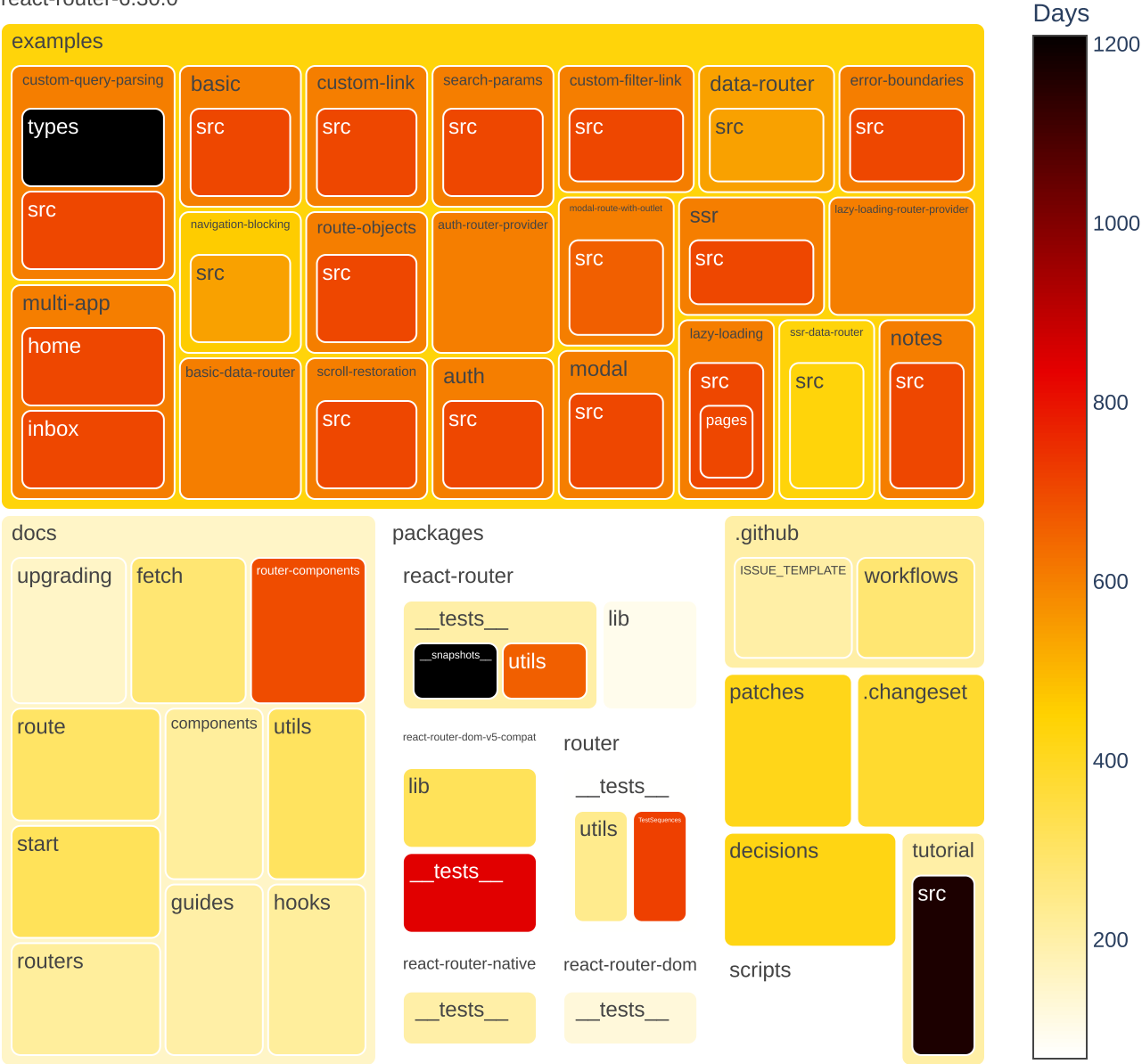
react-router-6.30.0



Days since last file modification per directory

Days since last file modification

react-router-6.30.0



Days since last file modification per directory (ranked)

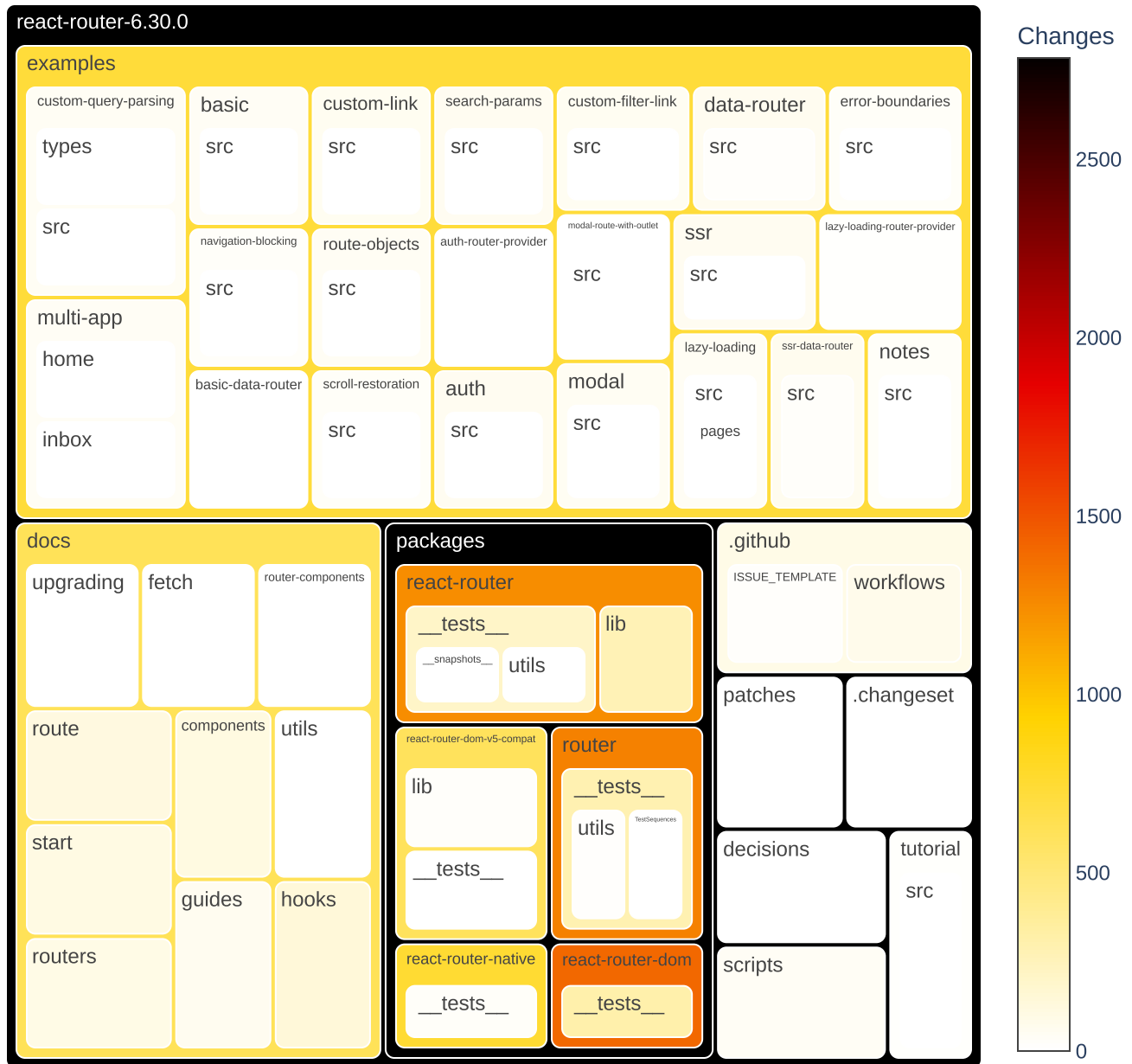
Rank of days since last file modification

react-router-6.30.0



File changed frequently with other files

Pairwise file changes



Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

Sum of commits that changed more than 30 files (each) = 229

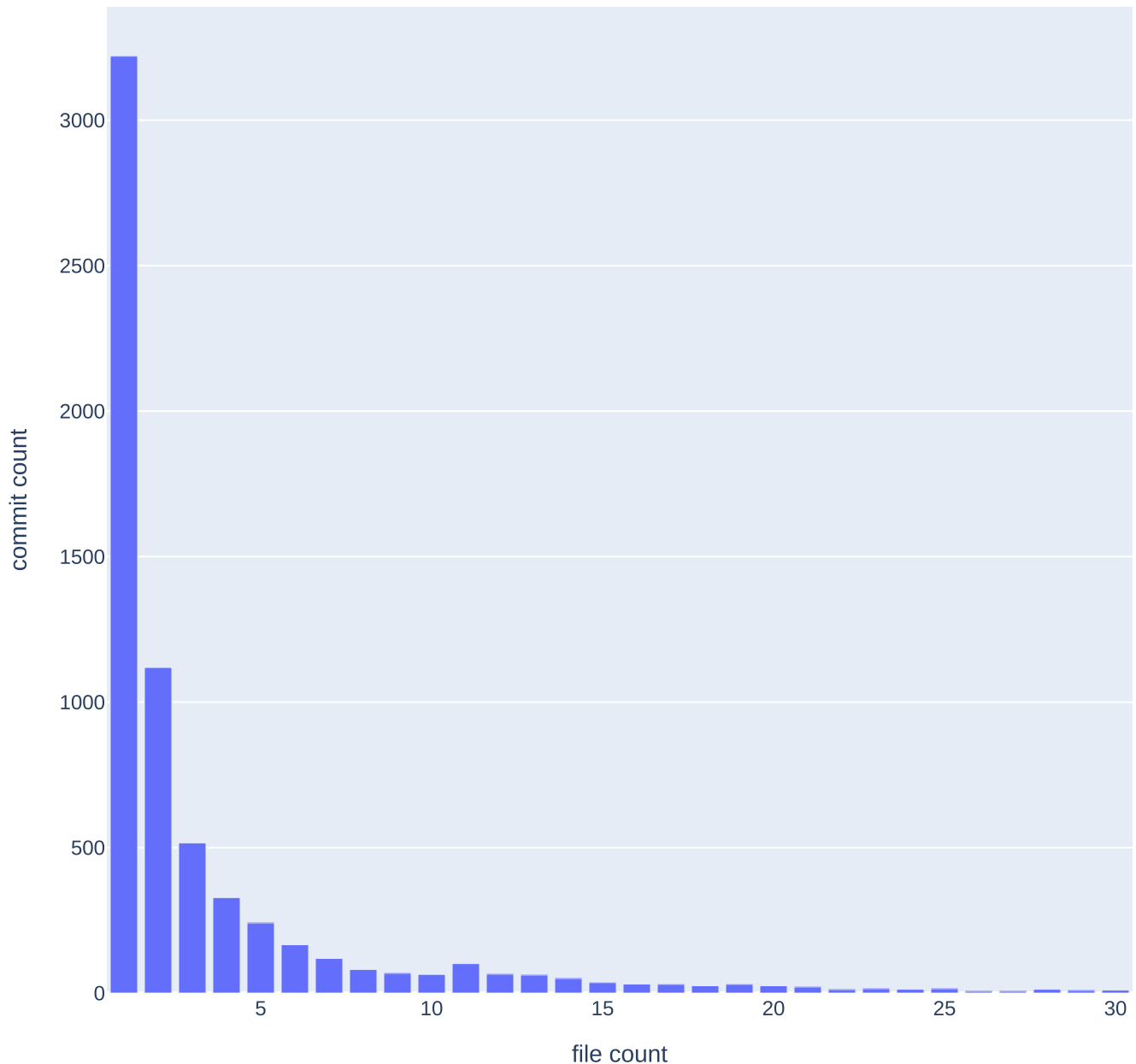
Max changed files with one commit = 498

	filesPerCommit	commitCount
count	117.000000	117.000000
mean	79.179487	57.487179
std	78.464424	318.285158
min	1.000000	1.000000
25%	30.000000	1.000000
50%	59.000000	3.000000
75%	96.000000	8.000000
max	498.000000	3220.000000

	filesPerCommit	commitCount
0	1	3220
1	2	1119
2	3	516
3	4	328
4	5	241
5	6	166
6	7	119
7	8	81
8	9	68
9	10	64
10	11	101
11	12	65
12	13	62
13	14	50
14	15	36
15	16	31
16	17	30
17	18	25
18	19	30
19	20	25
20	21	21
21	22	12
22	23	15
23	24	13
24	25	15
25	26	6
26	27	6
27	28	13
28	29	9
29	30	10

Bar chart with the number of files per commit distribution

Changed files per commit



Pairwise Changed Files vs. Dependency Weight

This section explores the correlation between how often pairs of files are changed together (common commit count) and their dependency weight. Note that these results should be interpreted cautiously, as comparing pairwise changes and dependencies is inherently challenging.

Considerations

- **Historical vs. Current State:** Pairwise changes reflect the entire git history, while dependency weight represents the current state of the codebase.

- **Commit Granularity:** Developers may use different commit strategies, such as squashing changes into a single commit or creating fine-grained commits. Ideally, each commit should represent a single semantic change for accurate analysis.
- **Dependency Representation:** Some file types (e.g., Java files with import statements) clearly define dependencies, while others (e.g., shell scripts, XML, YAML) lack explicit dependency relationships.
- **Repository Characteristics:** Repositories with generated code may have many large commits, while stabilized repositories may only update configuration files for dependency changes.

Data Preview

```
Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: weight)} {position: line: 9, column: 28, offset: 549} for query: "// List pair of files that were changed together and that have a declared dependency between each other.\n\n MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n //De-duplicating the pairs of files isn't necessary, because the dependency relation is directed.\n //WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n WITH firstCodeFile.fileName AS firstFileName\n      ,secondCodeFile.fileName AS secondFileName\n      ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n      ,pairwiseChange.commitCount AS commitCount\n      ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\n RETURN dependencyWeight\n      ,commitCount\n      ,fileDistanceAsFewestChangeDirectoryCommands\n      // ,count(*) AS occurrences\n // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\n ORDER BY dependencyWeight, commitCount\n\n\n // MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n // MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n // WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n\n // RETURN firstCodeFile.fileName AS firstFileName\n //      ,secondCodeFile.fileName AS secondFileName\n //      ,dependency.weight AS dependencyWeight\n //      ,pairwiseChange.commitCount AS commitCount\n // ORDER BY dependencyWeight, commitCount\n\n\n // MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n\n // WITH count(DISTINCT relation) AS relatedFilesCount\n // ,collect(DISTINCT relation) AS relations\n // UNWIND relations AS relation\n // WITH relatedFilesCount\n //      ,coalesce(relation.commitCount, 0) AS commitCount\n //      ,coalesce(relation.weight, 0) AS dependencyWeight\n //      ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n // RETURN dependencyWeight\n //      ,commitCount\n //      ,fileDistanceAsFewestChangeDirectoryCommands\n // ORDER BY dependencyWeight, commitCount\n"
```

```
Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.UnknownPropertyKeyWarning} {category: UNRECOGNIZED} {title: The provided property key is not in the database} {description: One of the property names in your query is not available in the database, make sure you didn't misspell it or that the label is available when you run this statement in your application (the missing property name is: fileDistanceAsFewestChangeDirectoryCommands)} {position: line: 11, column: 19, offset: 672} for query: "// List pair of files that were changed together and that have a declared dependency between each other.\n\nMATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\nMATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n//De-duplicating the pairs of files is n't necessary, because the dependency relation is directed.\n//WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\nWITH firstCodeFile.fileName AS firstFileName\n      ,secondCodeFile.fileName AS secondFileName\n      ,coalesce(dependency.weight, dependency.cardinality) AS dependencyWeight\n      ,pairwiseChange.commitCount AS commitCount\n      ,dependency.fileDistanceAsFewestChangeDirectoryCommands AS fileDistanceAsFewestChangeDirectoryCommands\nRETURN dependencyWeight\n      ,commitCount\n      ,fileDistanceAsFewestChangeDirectoryCommands\n      // ,count(*) AS occurrences\n      // ,collect(firstFileName + ' -> ' + secondFileName)[0..3] AS examples\nORDER BY dependencyWeight, commitCount\n\n// MATCH (firstCodeFile:File)-[dependency:DEPENDS_ON]->(secondCodeFile:File)\n// MATCH (firstCodeFile)-[pairwiseChange:CHANGED_TOGETHER_WITH]-(secondCodeFile)\n// WHERE elementId(firstCodeFile) < elementId(secondCodeFile)\n// RETURN firstCodeFile.fileName AS firstFileName\n//      ,secondCodeFile.fileName AS secondFileName\n//      ,dependency.weight AS dependencyWeight\n//      ,pairwiseChange.commitCount AS commitCount\n// ORDER BY dependencyWeight, commitCount\n\n// MATCH (g1:!Git&File)-[relation:CHANGED_TOGETHER_WITH|DEPENDS_ON]-(g2:!Git&File)\n// WITH count(DISTINCT relation) AS relatedFilesCount\n//      ,collect(DISTINCT relation) AS relations\n// UNWIND relations AS relation\n// WITH relatedFilesCount\n//      ,coalesce(relation.commitCount, 0) AS commitCount\n//      ,coalesce(relation.weight, 0) AS dependencyWeight\n//      ,coalesce(relation.fileDistanceAsFewestChangeDirectoryCommands, 0) AS fileDistanceAsFewestChangeDirectoryCommands\n// RETURN dependencyWeight\n//      ,commitCount\n//      ,fileDistanceAsFewestChangeDirectoryCommands\n// ORDER BY dependencyWeight, commitCount"
```

dependencyWeight commitCount fileDistanceAsFewestChangeDirectoryCommands

Data Statistics

'Pairwise changed git files compared to dependency weights - Overall statistics'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
count	0	0	0
unique	0	0	0
top	NaN	NaN	NaN
freq	NaN	NaN	NaN

'Pairwise changed git files compared to dependency weights - Pearson Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	NaN	NaN	NaN
commitCount	NaN	NaN	NaN
fileDistanceAsFewestChangeDirectoryCommands	NaN	NaN	NaN

'Pairwise changed git files compared to dependency weights - Spearman Correlation'

	dependencyWeight	commitCount	fileDistanceAsFewestChangeDirectoryCommands
dependencyWeight	NaN	NaN	NaN
commitCount	NaN	NaN	NaN
fileDistanceAsFewestChangeDirectoryCommands	NaN	NaN	NaN

Less than 5 samples are not enough to calculate p-values

No data to plot

WordCloud of git authors

	word	frequency
0	Michael Jackson	1832
1	Ryan Florence	1119
2	Matt Brophy	753
3	Jimmy Jia	381
4	Tim Dorr	326
5	Chance Strickland	180
6	Mateusz Zatorski	124
7	Logan McAnsh	99
8	Dan Abramov	92
9	Paul Sherman	77

Wordcloud of git authors

